# Dummy Variables and Interaction

# Outline

1. Qualitative regressors
2. Indicator variables
3. Inference on the indicator coefficient
4. Factors with more than two levels
5. Interaction terms

# Qualitative regressors

# Qualitative regressors

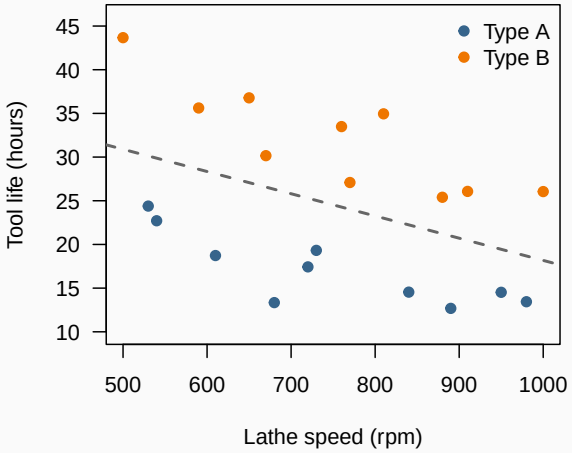
- So far every regressor has been quantitative: speed, temperature, delivery volume.
- Many important predictors are qualitative (categorical): sex, region, treatment group, machine type.
- A category has no numerical value, so it cannot enter $\mathbf{X}$ as it is.
- **Question:** how can we put “tool type A or B” into a regression model?

# Example dataset: tool life data (MPV Example 8.1)

- $y$: hours of tool life; $x_1$: lathe speed (rpm); two tool types, A and B, $n = 20$.

| Tool type A | | | Tool type B | | |
|---|---|---|---|---|---|
| $i$ | $y$ | $x_1$ | $i$ | $y$ | $x_1$ |
| 1 | 18.73 | 610 | 11 | 30.16 | 670 |
| 2 | 14.52 | 950 | 12 | 27.09 | 770 |
| 3 | 17.43 | 720 | 13 | 25.40 | 880 |
| 4 | 14.54 | 840 | 14 | 26.05 | 1000 |
| 5 | 13.44 | 980 | 15 | 33.49 | 760 |
| 6 | 24.39 | 530 | 16 | 35.62 | 590 |
| 7 | 13.34 | 680 | 17 | 26.07 | 910 |
| 8 | 22.71 | 540 | 18 | 36.78 | 650 |
| 9 | 12.68 | 890 | 19 | 34.95 | 810 |
| 10 | 19.32 | 730 | 20 | 43.67 | 500 |

# What happens if we ignore tool type?




- Simple linear regression of $y$ on $x_1$ (dashed): $\hat{y} = 43.62 - 0.0254\, x_1$.
- $R^2 = 0.186$, $\hat{\sigma} = 8.44$.
- The points split into two clusters: type B tools last longer at every speed.
- Tool type is a missing regressor; the residuals carry its effect.

# Indicator variables

# Indicator (dummy) variable

- An indicator variable encodes membership in a category with the values $0$ and $1$.
- For tool type:

$$x_2 = \begin{cases} 0 & \text{if the observation is from tool type A,} \\ 1 & \text{if the observation is from tool type B.} \end{cases}$$

- Type A is coded $0$: it is the **reference** (baseline) category.

# Tool life data with the indicator column

| Tool type A | | | | Tool type B | | | |
|---|---|---|---|---|---|---|---|
| $i$ | $y$ | $x_1$ | $x_2$ | $i$ | $y$ | $x_1$ | $x_2$ |
| 1 | 18.73 | 610 | 0 | 11 | 30.16 | 670 | 1 |
| 2 | 14.52 | 950 | 0 | 12 | 27.09 | 770 | 1 |
| 3 | 17.43 | 720 | 0 | 13 | 25.40 | 880 | 1 |
| 4 | 14.54 | 840 | 0 | 14 | 26.05 | 1000 | 1 |
| 5 | 13.44 | 980 | 0 | 15 | 33.49 | 760 | 1 |
| 6 | 24.39 | 530 | 0 | 16 | 35.62 | 590 | 1 |
| 7 | 13.34 | 680 | 0 | 17 | 26.07 | 910 | 1 |
| 8 | 22.71 | 540 | 0 | 18 | 36.78 | 650 | 1 |
| 9 | 12.68 | 890 | 0 | 19 | 34.95 | 810 | 1 |
| 10 | 19.32 | 730 | 0 | 20 | 43.67 | 500 | 1 |

$x_2 = 0$ for tool type A and $x_2 = 1$ for tool type B; $x_2$ is a new column of $\mathbf{X}$ next to $\mathbf{1}$ and $x_1$.

# Model with one quantitative regressor and one indicator

- Regress $y$ on speed $x_1$ and the indicator $x_2$:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2).$$

- This is an ordinary multiple linear regression; $\mathbf{X}$ has columns $\mathbf{1}$, $\mathbf{x}_1$, $\mathbf{x}_2$, and OLS applies unchanged.
- Only the interpretation of the coefficient on $x_2$ is new.

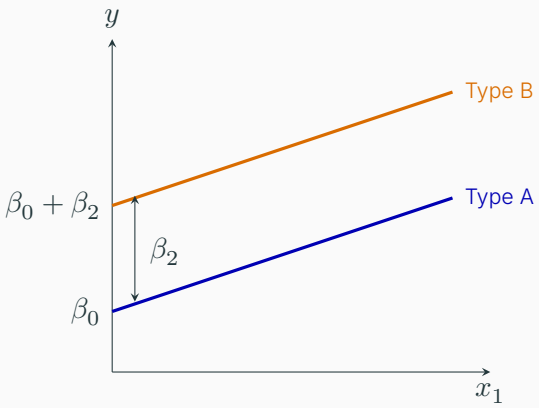
# Two regression lines from one model

- Setting $x_2 = 0$ and $x_2 = 1$ gives the mean response for each tool type:

$$\begin{aligned} &\text{Type A:} \quad \mathrm{E}[y \mid x_1, x_2 = 0] = \beta_0 + \beta_1 x_1, \\ &\text{Type B:} \quad \mathrm{E}[y \mid x_1, x_2 = 1] = (\beta_0 + \beta_2) + \beta_1 x_1. \end{aligned}$$

- The lines have the same slope $\beta_1$ and intercepts that differ by $\beta_2$: the lines are **parallel**.
- $\beta_2$ is the difference in mean tool life between type B and type A at any fixed speed.

# Geometry: parallel lines




- The indicator shifts the intercept only.
- The effect of the category does not depend on $x_1$: the vertical gap is $\beta_2$ everywhere.
- This is a strong assumption; we will test it with an interaction term.

# Tool life: fitted parallel-lines model

$$\hat{y} = 36.986 - 0.0266\,x_1 + 15.004\,x_2.$$

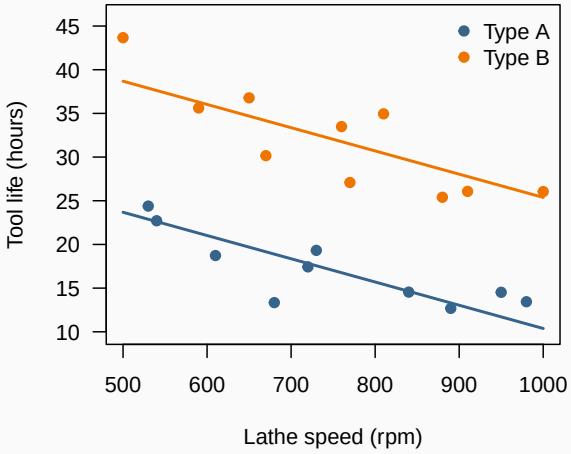
| Predictor | Estimate | Std. error | $t$ |
|---|---|---|---|
| Intercept | $36.986$ | $3.510$ | $10.54$ |
| $x_1$ (speed) | $-0.0266$ | $0.0045$ | $-5.89$ |
| $x_2$ (type B) | $15.004$ | $1.360$ | $11.04$ |

$R^2 = 0.900$, $\quad \hat{\sigma} = 3.04$ on 17 degrees of freedom.

# Interpreting the coefficients

- **Question:** what do $\hat{\beta}_1 = -0.0266$ and $\hat{\beta}_2 = 15.004$ mean in this problem?

# Fitted lines and data




- Type A: $\hat{y} = 36.99 - 0.0266\, x_1$.
- Type B: $\hat{y} = 51.99 - 0.0266\, x_1$.
- Adding the indicator raises $R^2$ from $0.186$ to $0.900$ and cuts $\hat{\sigma}$ from $8.44$ to $3.04$.

# Inference on the indicator coefficient

# Full and reduced models

- A hypothesis $H_0$ about the coefficients, e.g. that some of them are zero, defines a simpler model.
- **Full model (FM):** all terms; $p$ parameters.
- **Reduced model (RM):** the FM with $H_0$ imposed; $p - q$ parameters, where $q$ is the number of restrictions in $H_0$.
- The RM is *nested* in the FM: every RM is a special case of the FM.

Example: $H_0 : \beta_2 = 0$ ($q = 1$).

$$\begin{aligned} \text{FM}: &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon, \\ \text{RM}: &\quad y = \beta_0 + \beta_1 x_1 + \varepsilon. \end{aligned}$$

# Extra sum of squares

- Least squares searches over all coefficient values allowed by the model; the FM allows more, so
$$\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) \geq \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}).$$

- The difference is the **extra sum of squares**: the increase in the residual sum of squares caused by imposing $H_0$,
$$\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) \geq 0.$$

- Small increase: the dropped terms add little, and $H_0$ is plausible. Large increase: the dropped terms matter.
- **Question:** how large is “large”?

# $F$-test for nested models

- Under $H_0$, with $n$ observations, $p$ parameters in the FM and $q$ restrictions:

$$F_0 = \frac{\left[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})\right]/q}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n-p)} \sim F_{q,\,n-p}.$$

- Numerator: increase in $\mathrm{SS}_{\mathrm{Res}}$ per restriction. Denominator: $\mathrm{MS}_{\mathrm{Res}}$ of the FM, an estimate of $\sigma^2$.
- We use this distributional result here; Lecture 9 derives it.
- Reject $H_0$ at level $\alpha$ if $F_0 > F_{\alpha,\,q,\,n-p}$.
- For $q = 1$ the test is equivalent to the two-sided $t$-test of the coefficient, with $F_0 = t_0^2$.

# Inference on the indicator coefficient

- $H_0 : \beta_2 = 0$ (the two tool types have the same mean life at every speed) is tested with the usual $t$-test; here $t_0 = 11.04$ on 17 df.
- Equivalently, use the $F$-test with FM: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ and RM: $y = \beta_0 + \beta_1 x_1 + \varepsilon$ (speed only); here $q = 1$, $n - p = 17$:

$$F_0 = \frac{\left[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})\right]/1}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/17} = \frac{(1282.08 - 157.05)/1}{157.05/17} = 121.8,$$

$$F_0 = t_0^2.$$

- Reject $H_0$: tool type is strongly associated with tool life after adjusting for speed.

# Special case: indicator only = two-sample $t$-test

- Drop speed and regress $y$ on the indicator alone: $y = \beta_0 + \beta_2 x_2 + \varepsilon$.
- The group means are $\mathrm{E}[y \mid x_2 = 0] = \beta_0$ and $\mathrm{E}[y \mid x_2 = 1] = \beta_0 + \beta_2$, and OLS gives the sample means:

$$\hat{\beta}_0 = \bar{y}_A = 17.11, \qquad \hat{\beta}_2 = \bar{y}_B - \bar{y}_A = 31.93 - 17.11 = 14.82.$$

- $t_0 = \hat{\beta}_2 / \operatorname{se}(\hat{\beta}_2) = 14.82/2.30 = 6.44$ on $n - 2 = 18$ df: exactly the pooled-variance two-sample $t$-test of $\mu_A = \mu_B$.
- With $m$ levels and $m - 1$ indicators, the overall $F$-test of the regression is the one-way ANOVA $F$-test.
- Adding speed raises $t_0$ for tool type from $6.44$ to $11.04$: speed explains part of the noise, so $\hat{\sigma}$ falls from $5.15$ to $3.04$.

# Factors with more than two levels

# Categorical regressor with $m$ levels

- A categorical regressor is called a **factor**; its categories are its **levels** (tool type is a factor with levels A and B).
- We write $m$ for the number of levels ($k$ remains the number of regressors, $p = k + 1$).
- A factor with $m$ levels is represented by $m - 1$ indicator variables.
- The level without an indicator is the reference level.
- Suppose there were three tool types A, B, C (hypothetical extension), with A as reference:

$$x_2 = \begin{cases} 1 & \text{if type B,} \\ 0 & \text{otherwise,} \end{cases} \qquad x_3 = \begin{cases} 1 & \text{if type C,} \\ 0 & \text{otherwise.} \end{cases}$$

# Coding and interpretation for $m = 3$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

| Type | $x_2$ | $x_3$ | $\mathrm{E}[y \mid x_1, x_2, x_3]$ |
|---|---|---|---|
| A | 0 | 0 | $\beta_0 + \beta_1 x_1$ |
| B | 1 | 0 | $(\beta_0 + \beta_2) + \beta_1 x_1$ |
| C | 0 | 1 | $(\beta_0 + \beta_3) + \beta_1 x_1$ |

$\beta_2$ and $\beta_3$ are the differences of B and C from the reference A, at fixed $x_1$. The difference between B and C is $\beta_2 - \beta_3$.

# The dummy variable trap

- Suppose we include all $m = 3$ indicators $\mathbf{d}_A, \mathbf{d}_B, \mathbf{d}_C$ together with an intercept, where $\mathbf{d}_B = \mathbf{x}_2$, $\mathbf{d}_C = \mathbf{x}_3$ and $\mathbf{d}_A$ is the indicator of type A.
- Every observation belongs to exactly one type, so

$$\mathbf{1} = \mathbf{d}_A + \mathbf{d}_B + \mathbf{d}_C,$$
$$\operatorname{rank}(\mathbf{X}) = 4 < 5 \text{ columns.}$$

- $\mathbf{X}$ lacks full column rank, so $\mathbf{X}'\mathbf{X}$ is singular (Lecture 4): the normal equations have no unique solution and $\boldsymbol{\beta}$ is not **identifiable**, i.e. different coefficient vectors give the same $\mathrm{E}[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta}$.
- Remedy: drop one indicator (use $m - 1$), or drop the intercept (then each coefficient is a level-specific intercept).

# Question: counting indicators

- A factor has $4$ levels and the model includes an intercept. How many indicators are needed? How many if the intercept is removed, and how does the meaning of the coefficients change?

# Choice of reference level

- Changing the reference level changes the coefficients, not the model.
- With reference C, use the indicators $d_A$, $d_B$:
  $\mathrm{E}[y] = \beta'_0 + \beta'_1 x_1 + \beta'_A d_A + \beta'_B d_B$.
- Both codings describe the same three lines, so the mean of each type must agree:

| Type | Reference A | Reference C | Solve |
|---|---|---|---|
| C | $(\beta_0 + \beta_3) + \beta_1 x_1$ | $\beta'_0 + \beta'_1 x_1$ | $\beta'_0 = \beta_0 + \beta_3,\ \beta'_1 = \beta_1$ |
| A | $\beta_0 + \beta_1 x_1$ | $(\beta'_0 + \beta'_A) + \beta'_1 x_1$ | $\beta'_A = \beta_0 - \beta'_0 = -\beta_3$ |
| B | $(\beta_0 + \beta_2) + \beta_1 x_1$ | $(\beta'_0 + \beta'_B) + \beta'_1 x_1$ | $\beta'_B = \beta_2 - \beta_3$ |

- The slope $\beta_1$ does not change; only the differences are measured from C instead of A.

# Reference level: what changes and what does not

- Coefficients of the indicators and the intercept change with the reference level; their $t$-tests compare each level with a different baseline.
- Fitted values, residuals, $\mathrm{SS}_{\mathrm{Res}}$, $R^2$ and the overall $F$-test of the factor are identical for every choice of reference level.
- Choose the reference level that makes the comparisons of interest direct (e.g. a control group).

# Testing the factor as a whole

- Individual $t$-tests compare one level with the reference only.
- To test whether the factor matters at all, $H_0 : \beta_2 = \beta_3 = 0$, use the partial $F$-test:

$$F_0 = \frac{\left[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})\right]/(m-1)}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n-p)} \sim F_{m-1,\,n-p} \quad \text{under } H_0,$$

- where the reduced model omits all $m-1$ indicators and $p$ is the number of parameters of the full model.

# Indicator variables versus allocated codes

- Tempting alternative: one regressor with codes A $= 1$, B $= 2$, C $= 3$ (allocated codes).
- This forces $\mathrm{E}[y]$ to change by the same amount $\beta$ from A to B and from B to C, as if the levels were equally spaced on a numeric scale.
- For **nominal** categories (no natural order, e.g. tool type or region) that ordering and spacing is arbitrary; recoding to A $= 1$, C $= 2$, B $= 3$ would change the fit.
- Indicator variables impose no such structure: each level gets its own mean shift.
- Allocated codes are defensible only when the levels are genuinely ordered and roughly equally spaced.

# Interaction terms

# Do both tool types respond to speed in the same way?

- The parallel-lines model assumes that increasing speed shortens life by the same amount for tool types A and B.
- **Question:** how can we let the slope on $x_1$ differ between the two types?

# Indicator $\times$ quantitative interaction

- Add the product of the indicator and the quantitative regressor:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_{12}\, x_{i1} x_{i2} + \varepsilon_i.$$

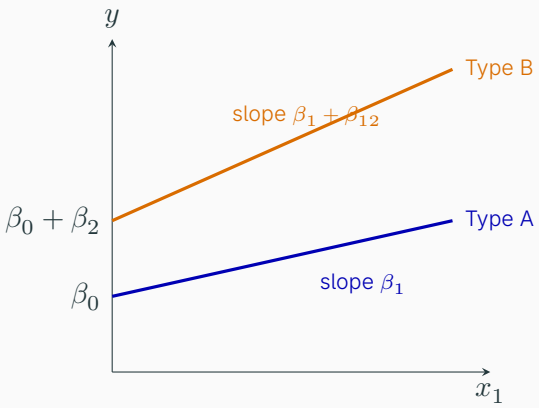
- The subscript of $\beta_{12}$ (read “beta one-two”, not “beta twelve”) names the regressors whose product the term contains: $x_1$ and $x_2$. In general, the coefficient of $x_j x_l$ is written $\beta_{jl}$.
- The effect of $x_1$ now depends on the category, and the effect of the category depends on $x_1$.

# Group-specific lines

$$\begin{aligned}
&\text{Type A:} \quad \mathrm{E}[y \mid x_1, x_2 = 0] = \beta_0 + \beta_1 x_1, \\
&\text{Type B:} \quad \mathrm{E}[y \mid x_1, x_2 = 1] = (\beta_0 + \beta_2) + (\beta_1 + \beta_{12})\, x_1.
\end{aligned}$$

- $\beta_2$: difference in intercepts; $\beta_{12}$: difference in slopes.
- The difference between the types at speed $x_1$ is $\beta_2 + \beta_{12} x_1$; it changes linearly with $x_1$.

# Geometry: non-parallel lines




- Two free intercepts and two free slopes.
- The difference B $-$ A grows with $x_1$ when $\beta_{12} > 0$.
- If $\beta_{12} \neq 0$, the lines cross at $x_1 = -\beta_2/\beta_{12}$ (if this lies in the range of the data, the ranking of the types reverses).

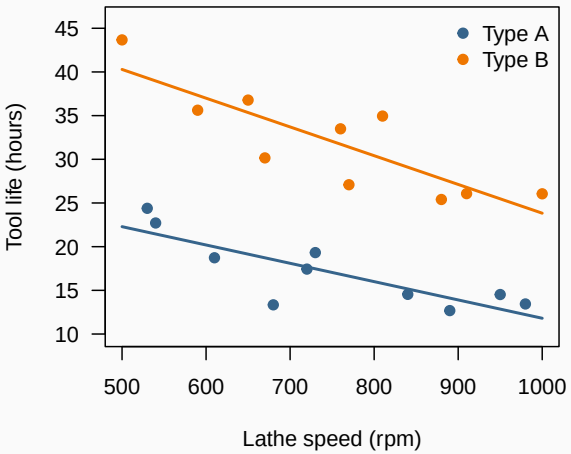
# Tool life: fitted interaction model

$$\hat{y} = 32.775 - 0.0210\,x_1 + 23.971\,x_2 - 0.0119\,x_1 x_2.$$

| Predictor | Estimate | Std. error | $t$ |
|---|---|---|---|
| Intercept | $32.775$ | $4.634$ | $7.07$ |
| $x_1$ (speed) | $-0.0210$ | $0.0061$ | $-3.45$ |
| $x_2$ (type B) | $23.971$ | $6.769$ | $3.54$ |
| $x_1 x_2$ ($\beta_{12}$) | $-0.0119$ | $0.0088$ | $-1.35$ |

$R^2 = 0.911$, $\hat{\sigma} = 2.97$ on $16$ degrees of freedom.

# Tool life: different slopes by tool type




- Type A: $\hat{y} = 32.78 - 0.0210\, x_1$.
- Type B: $\hat{y} = 56.75 - 0.0329\, x_1$.
- Steeper slope for type B ($\hat{\beta}_{12} = -0.0119$): the lines converge as speed increases.
- They would cross at $-\hat{\beta}_2/\hat{\beta}_{12} \approx 2007$ rpm, far beyond the data (500–1000 rpm).
- Is the slope difference larger than sampling variation?

# Testing the interaction

- $H_0 : \beta_{12} = 0$ (parallel lines): is the interaction term needed beyond $x_1$ and $x_2$? Compare nested models, with $q = 1$ and $n - p = 16$:

$$\begin{aligned} \text{FM}: &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon, \\ \text{RM}: &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon. \end{aligned}$$

# Testing the interaction: result

$$F_0 = \frac{\left[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})\right]/1}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/16} = \frac{(157.05 - 140.98)/1}{140.98/16} = 1.82,$$

$$t_0 = -1.35 \quad (t_0^2 = F_0), \qquad p\text{-value} = 0.196.$$

- No evidence that the slopes differ: the parallel-lines model is adequate for these data.
- The estimated slopes differ by $0.0119$ per rpm, but with $n = 20$ the estimate is too imprecise to distinguish it from zero.

# Testing sequence

- Test the interaction first: the meaning of $\beta_2$ depends on whether $x_1x_2$ is in the model, so the indicator is tested in the model that survives.

| Step | $H_0$ | Question | FM vs. RM | $F_0$ | $p$ |
|---|---|---|---|---|---|
| 1 | $\beta_{12} = 0$ | Parallel lines? | interaction vs. parallel | $1.82$ | $0.196$ |
| 2 | $\beta_2 = 0$ | Do they coincide? | parallel vs. speed only | $121.8$ | $< 0.001$ |

- Step 1 does not reject, so $x_1x_2$ is dropped and step 2 is carried out in the parallel-lines model.
- Step 2 rejects. Conclusion for tool life: keep $x_1$ and $x_2$, drop $x_1x_2$.

# One-step alternative: are the two lines identical?

- Test both differences at once, $H_0 : \beta_2 = \beta_{12} = 0$ ($q = 2$): FM is the interaction model, RM is the speed-only regression:

$$F_0 = \frac{(1282.08 - 140.98)/2}{140.98/16} = 64.8 \quad (p < 0.001).$$

- Reject: one common line is not adequate; tool type matters.
- Rejection does not say whether the intercepts, the slopes or both differ; the two-step sequence answers that.

# Question: why not fit two separate regressions?

- The interaction model gives one line for each tool type. Why not simply fit two regressions, one for each type?

# Indicator variables versus separate regressions

- Separate fits (type A, type B) reproduce the interaction model coefficients:

| Fit | Intercept | Slope |
|---|---|---|
| Type A alone | $32.775 = \hat{\beta}_0$ | $-0.0210 = \hat{\beta}_1$ |
| Type B alone | $56.745 = \hat{\beta}_0 + \hat{\beta}_2$ | $-0.0329 = \hat{\beta}_1 + \hat{\beta}_{12}$ |

- Only the single model allows direct tests of $\beta_2$, $\beta_{12}$, and of the special cases (parallel, equal lines).

# Common versus separate error variance

- The interaction model assumes a common $\sigma^2$ and estimates it from all $n$ observations; separate fits estimate one $\sigma^2$ per group.

| Fit | $\mathrm{SS}_{\mathrm{Res}}$ | df | $\hat{\sigma}^2 = \mathrm{MS}_{\mathrm{Res}}$ |
|---|---|---|---|
| Type A alone | $47.57$ | $8$ | $5.95$ |
| Type B alone | $93.40$ | $8$ | $11.68$ |
| Interaction model (pooled) | $140.98$ | $16$ | $8.81$ |

- Pooled $\mathrm{SS}_{\mathrm{Res}}$ = sum of the separate ones; its $16$ df (instead of $8$) give shorter intervals and more powerful tests, if the groups share one $\sigma^2$.
- The group estimates differ by a factor of about $2$; with $8$ df each this is within sampling variation (variance-ratio $F$-test, $p \approx 0.36$). Check with residual plots by group (Lecture 6).

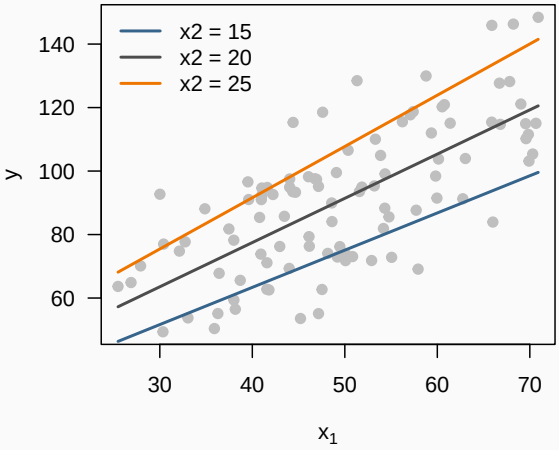
# Interaction between two quantitative regressors

- Interaction is not limited to indicators:
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon.$$

- The effect on $\mathrm{E}[y]$ of a unit change in $x_1$ depends on the level of $x_2$:
$$\frac{\partial\, \mathrm{E}[y]}{\partial x_1} = \beta_1 + \beta_{12}\, x_2, \qquad \frac{\partial\, \mathrm{E}[y]}{\partial x_2} = \beta_2 + \beta_{12}\, x_1.$$

- The mean response surface is no longer a plane; slices at fixed $x_2$ are lines with different slopes (this is the indicator case with $x_2 \in \{0, 1\}$).

# Two quantitative regressors: slices of the fitted surface




- Simulated data, $n = 100$: $\hat{y} = 0.69 + 0.508\, x_1 + 1.053\, x_2 + 0.0443\, x_1 x_2$.
- Fitted lines in $x_1$ at $x_2 = 15, 20, 25$ have slopes $\hat{\beta}_1 + \hat{\beta}_{12} x_2 = 1.17,\ 1.39,\ 1.61$.
- The larger $x_2$, the steeper the effect of $x_1$: the lines fan out.

# Interpreting coefficients when an interaction is present

- The coefficients of the single regressors ($\beta_1$, $\beta_2$) are called **main effects**, as opposed to the interaction $\beta_{12}$. With $x_1 x_2$ in the model, $\beta_1$ is the effect of $x_1$ only at $x_2 = 0$; it is not an overall effect of $x_1$ unless $x_2 = 0$ is meaningful.
- Centering the quantitative regressors at their means makes $\beta_1$ and $\beta_2$ interpretable at average values. It also reduces **collinearity**, i.e. strong correlation between $x_1 x_2$ and its components, which inflates standard errors (Lecture 4: nearly dependent columns make $\mathbf{X}'\mathbf{X}$ nearly singular; treated in detail in a later lecture).

# Practical guidance on interactions

- **Hierarchy principle:** whenever $x_1 x_2$ is in the model, keep the lower-order terms $x_1$ and $x_2$, even if their $t$-tests are not significant.
- Do not interpret interaction coefficients in isolation; plot the fitted lines at chosen values of the other regressor.
- Each additional interaction costs a degree of freedom; add them for substantive reasons, not by default.

# Interaction with an $m$-level factor

- For a three-level factor (indicators $x_2$, $x_3$) allow a separate slope for each level:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \varepsilon.$$

- Slopes in $x_1$: $\beta_1$ (reference A), $\beta_1 + \beta_{12}$ (B), $\beta_1 + \beta_{13}$ (C).
- Test of equal slopes, $H_0$: $\beta_{12} = \beta_{13} = 0$: partial $F$-test with $m - 1 = 2$ numerator degrees of freedom.

# Summary

- An indicator variable encodes a binary category as $0/1$; a factor with $m$ levels needs $m-1$ indicators plus an intercept.
- Indicators shift the intercept: they give parallel lines with common slope. Coefficients are differences from the reference level.
- Including all $m$ indicators with an intercept makes $\mathbf{X}'\mathbf{X}$ singular (dummy variable trap).
- The reference level changes coefficients, not the fit; test a whole factor with the partial $F$-test.
- Interaction terms $x_1 x_2$ let the slope depend on the category (or on another regressor); test them first with the extra-sum-of-squares $F$-test (RM versus FM).
- Tool life data: type matters ($F_0 = 121.8$); the slope difference is not significant ($p = 0.196$).

# Questions?